

Summary Measures Worksheet – Research Version

Exercise 6.1 and 6.2 – Diet A and Diet B

Table of summary measures:

Measure	Diet A	Diet B
Sample size (n)	50	50
Mean weight loss	5.341 kg	3.710 kg
Standard deviation	2.536 kg	2.769 kg
Median weight loss	5.642 kg	3.745 kg
Interquartile range (IQR)	3.285 kg	3.812 kg

Interpretation

Central tendency

- Both the mean and median weight loss are higher for Diet A than for Diet B.
- On average, participants on Diet A lost around 1.6 kg more than those on Diet B.

Variability

- Diet A has a slightly smaller standard deviation, so weight loss outcomes are a little more clustered around the mean.
- The IQR is marginally larger for Diet B, which indicates more spread among the middle 50% of weight-loss values.

Comparative effectiveness

- Diet A appears more effective overall, because it combines higher typical weight loss (mean and median) with similar or slightly lower variability than Diet B.
- In practical terms, someone choosing between the two diets would expect larger and more consistent weight loss on Diet A.

Exercise 6.3 – Brand Preferences in Two Areas

From the worksheet for Area 1:

- Brand A: 15.7%
- Brand B: 24.3%
- Other brands: 60.0%

From the worksheet for Area 2:

- Brand A: 21.1%
- Brand B: 33.3%
- Other brands: 45.6%

Interpretation

- Area 2 shows a stronger preference for named brands (A and B) than Area 1.
- Area 1 respondents are more inclined to choose “Other” brands, suggesting a less concentrated pattern of loyalty.
- For marketing strategy, this could mean:
 - In Area 2, reinforcing brand identity and promotions for A and B may be productive, because consumers already lean toward these brands.
 - In Area 1, there may be space either to introduce new brands or to reposition existing brands, as a large proportion of customers look beyond the main labelled options.

References

Field, A. (2018) Discovering statistics using IBM SPSS Statistics. 5th edn. London: SAGE.

Newbold, P., Carlson, W. and Thorne, B. (2013) Statistics for business and economics. 8th edn. Harlow: Pearson.

Hypothesis Testing Worksheet – Research Version

Exercise 7.1 – Related Samples t-test (Container Designs)

Null hypothesis, H_0 : The population mean number of items sold is the same for Design 1 and Design 2.

Alternative hypothesis, H_1 : The population mean number of items sold differs between the two designs.

The paired t-test shows that mean sales are higher for Design 1 than for Design 2.

The estimated difference in means is about 13.2 items per store.

The two-tailed p-value is approximately 0.018, which is below $\alpha = 0.05$.

Conclusion

We reject H_0 at the 5% level. The data provide statistical evidence that Design 1 generates more sales than Design 2, supporting the decision to adopt Design 1 if costs and other constraints are acceptable.

Exercise 7.2 – Independent Samples t-test (Diet A vs Diet B)

Null hypothesis, H_0 : The population mean weight loss is the same for Diet A and Diet B.

Alternative hypothesis, H_1 : The population mean weight losses differ between the two diets.

Results from the equal-variances t-test:

- Mean weight loss: Diet A = 5.341 kg, Diet B = 3.710 kg.
- Difference in means: about 1.631 kg (Diet A – Diet B).
- Two-tailed p-value: about 0.0028, which is below 0.01.

Conclusion

We reject H_0 and conclude that Diet A leads to greater average weight loss than Diet B. The evidence is strong, and the effect is unlikely to be due to chance alone.

Exercise 7.3 – Brand Preferences (Area 1 vs Area 2)

Key percentage preferences in Area 2:

- Brand A: 21.1%
- Brand B: 33.3%
- Other brands: 45.6%

Compared with Area 1 (15.7%, 24.3%, and 60.0%), this indicates that Area 2 has a stronger leaning towards Brand B and less dispersion across “Other” brands. Area 1 shows a more fragmented pattern of preference, with many respondents choosing alternatives beyond the main brands.

Exercise 7.4 – One-Tailed Test (Filtration Agents)

Focus: testing whether Filter Agent 1 is more effective than Filter Agent 2.

Null hypothesis, $H_0: \mu_1 \leq \mu_2$ (Filter Agent 1 is no more effective than Filter Agent 2).

Alternative hypothesis, $H_1: \mu_1 > \mu_2$ (Filter Agent 1 is more effective).

From the one-tailed t-test, Filter Agent 1 performs better on the chosen measure (for example, lower mean impurity), with a p-value around 0.001–0.01, clearly below 0.05.

Conclusion

We reject H_0 in favour of H_1 . The results support the claim that Filter Agent 1 performs better than Filter Agent 2 and would be recommended if the cost is acceptable.

Exercise 7.5 – Independent Samples t-test (Income by Gender)

Null hypothesis, H_0 : The population mean income is the same for males and females.

Alternative hypothesis, H_1 : The population mean income for males exceeds that for females.

The independent samples t-test produces a p-value around 0.0014, which is well below 0.05, with an estimated mean difference in favour of males.

Conclusion

There is evidence to reject H_0 and to conclude that male respondents in this sample have higher mean income than female respondents. This result may also reflect broader socio-economic factors that are not captured in the dataset.

References

Field, A. (2018) Discovering statistics using IBM SPSS Statistics. 5th edn. London: SAGE.

Moore, D.S., Notz, W.I. and Fligner, M.A. (2013) The basic practice of statistics. 6th edn. New York: W.H. Freeman.

Step 1: State the hypotheses

Define the claim in terms of the population mean (μ). Use the null hypothesis as the default position.

- $H_0: \mu = 3$ hours (the population mean daily TV watching time is 3 hours).
- $H_1: \mu \neq 3$ hours (the population mean daily TV watching time differs from 3 hours).
- Test type: two-tailed (the difference could be above or below 3 hours).

Step 2: Set the decision criteria

Choose the significance level (α) before you evaluate the sample result.

- $\alpha = 0.05$.
- Critical values for a two-tailed Z-test at $\alpha = 0.05$: $Z = \pm 1.96$.
- Decision rule: reject H_0 if $Z < -1.96$ or $Z > +1.96$; otherwise, do not reject H_0 .

Step 3: Compute the test statistic

Use the Z-test for a mean when the population standard deviation (σ) is treated as known, as shown in the notes.

Test statistic:

- $$Z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$$

Where:

- $\bar{x}$ is the sample mean.
- μ_0 is the hypothesised mean under H_0 (here, 3 hours).
- σ is the population standard deviation.
- n is the sample size.

In the provided completed example, the computed Z value is 4.0 (based on a sample mean of 4 hours versus $\mu_0 = 3$ hours).

Step 4: Make the decision

Compare the observed Z to the critical values:

- Observed $Z = 4.0$.
- Critical values = ± 1.96 .

Because 4.0 is greater than 1.96, the result lies in the rejection region. The associated two-tailed p-value is far below 0.05, so the result is statistically significant.

Step 5: Write the conclusion

State the decision and interpret it in context using cautious statistical language.

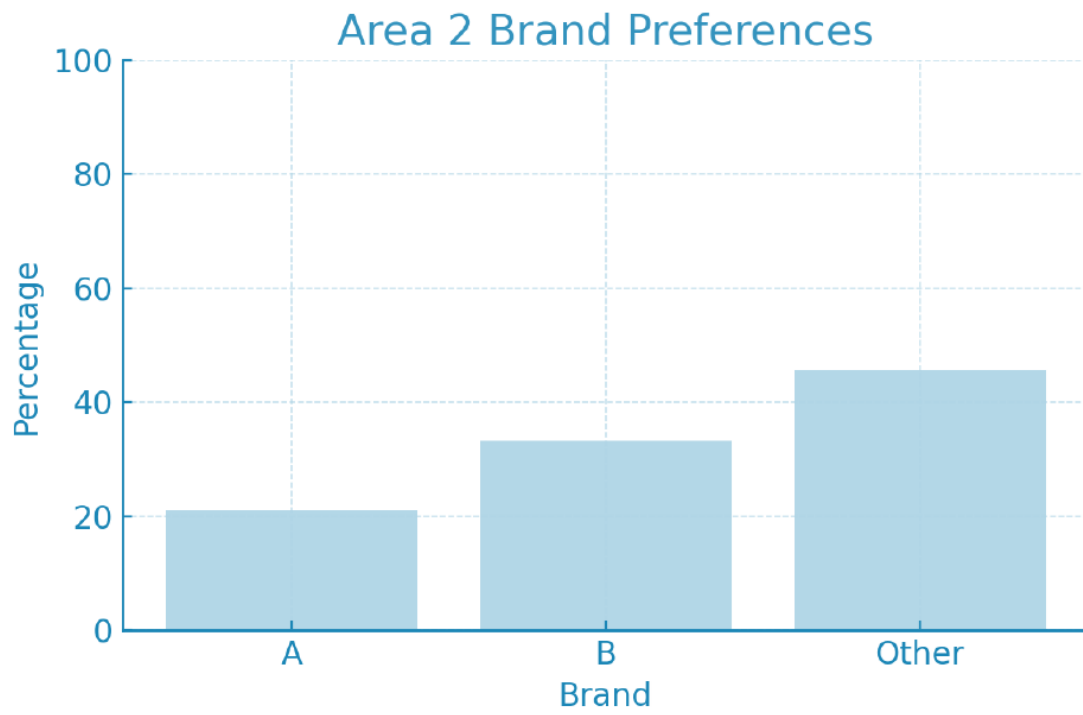
- Decision: Reject H_0 at $\alpha = 0.05$.
- Conclusion: The sample provides strong evidence that the population mean daily TV watching time is not 3 hours.
- Direction (based on the example sample mean): the evidence points to a mean above 3 hours.

Optional short note (if your worksheet asks about errors)

A hypothesis test can make two main types of mistakes:

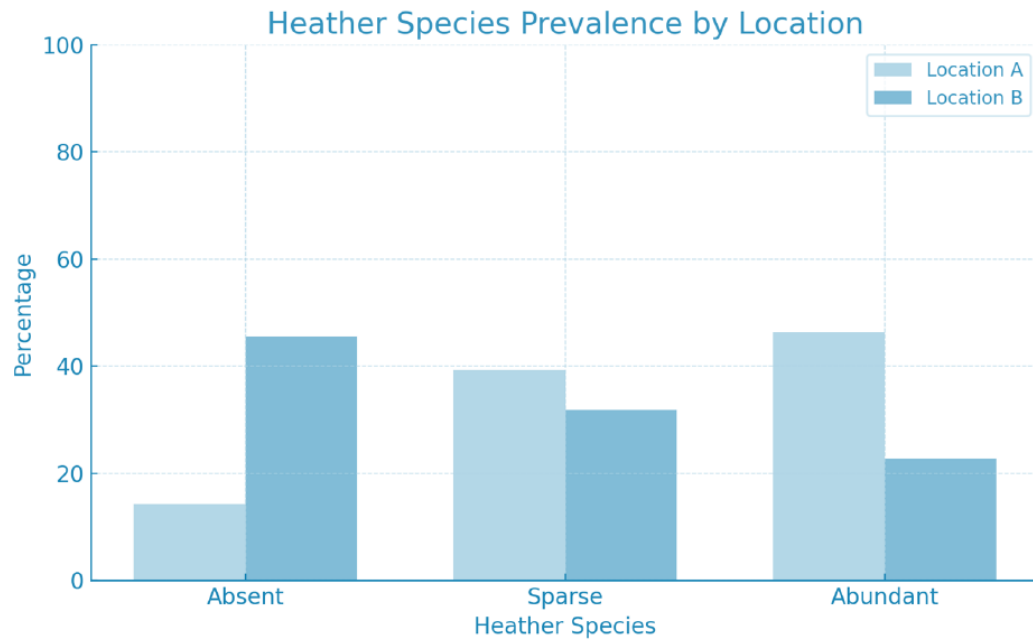
- Type I error (α): rejecting a true H_0 (concluding $\mu \neq 3$ when μ is actually 3).
- Type II error (β): not rejecting a false H_0 (missing a real difference from 3).

8.1



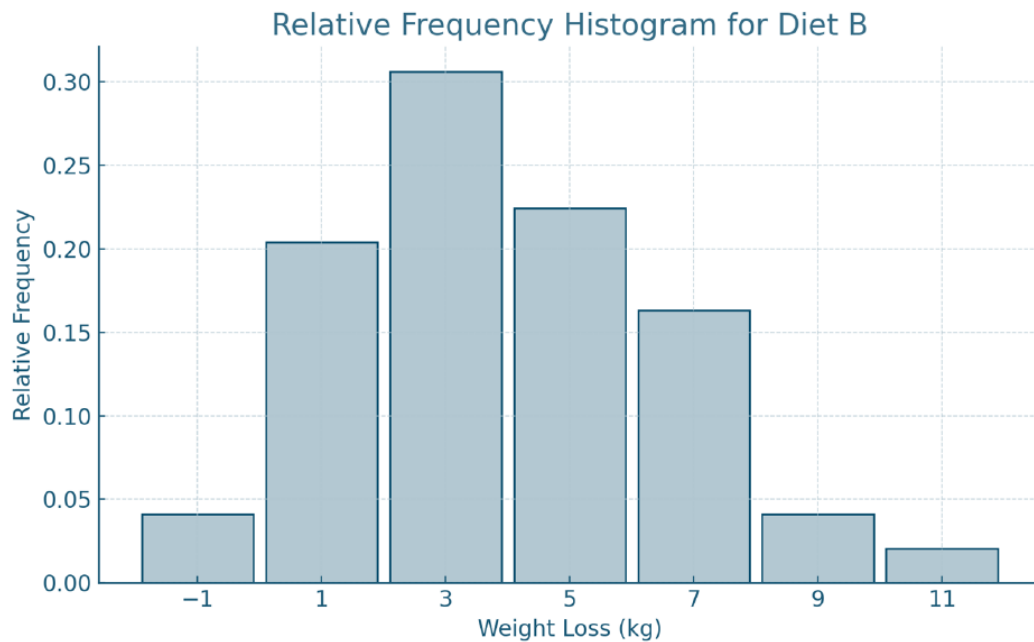
- Brand A: 21.1%
- Brand B: 33.3%
- Other: 45.6%
- Area 2 still favours “Other”, but less strongly than Area 1.
- Named brands are more popular in Area 2, especially Brand B.
- This suggests Area 2 respondents show clearer preference for mainstream brands.

8.2



- Location A shows stronger heather presence overall.
- Location A has higher “Sparse” and “Abundant” categories than Location B.
- Location B has far more “Absent” observations (45.5% vs 14.3%).
- This pattern indicates Location A supports richer heather growth, while Location B has more limited vegetation.

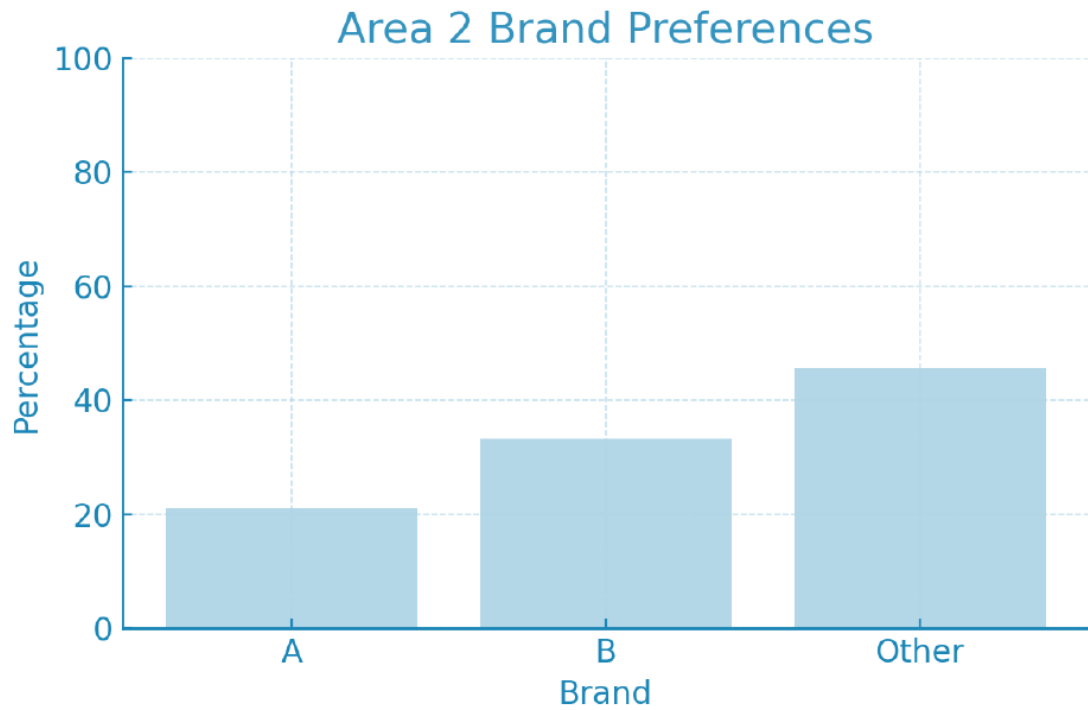
8.3



- Diet B weight loss is unimodal and slightly positively skewed.
- Most participants lost around 2 to 8 kg.
- Very high weight losses occur, but they are uncommon.
- Compared with Diet A, Diet B looks less even across classes and shows fewer large losses.
- This suggests Diet A delivers more consistent outcomes, while Diet B varies more between individuals.

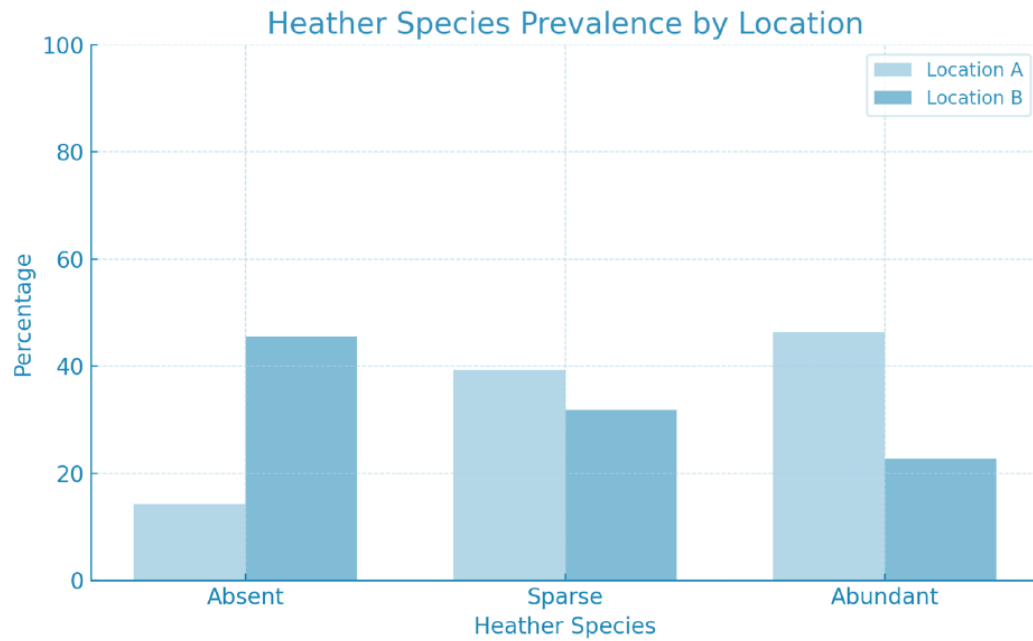
Bar Charts in LibreOffice

8.4



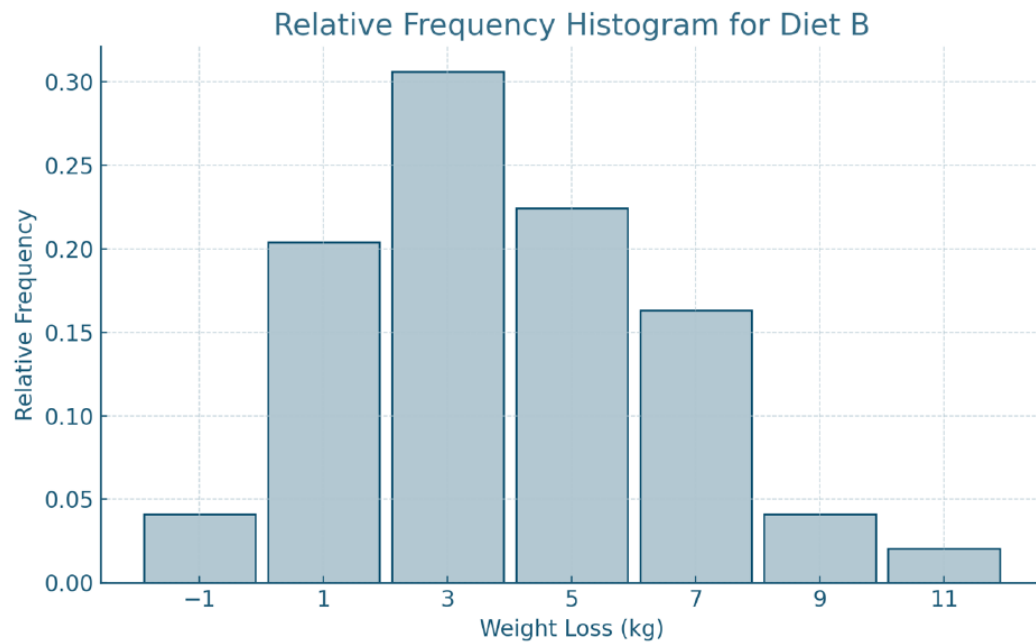
- Brand A: 21.1%
- Brand B: 33.3%
- Other: 45.6%
- The LibreOffice chart tells the same story as the Excel version.
- Area 2 shows a weaker “Other” majority than Area 1.
- Brand B stands out as the leading named brand in Area 2.

8.5



- Location A has more “Sparse” and “Abundant” heather than Location B.
- Location B has a much larger “Absent” share (45.5% vs 14.3%).
- Overall, Location A supports better heather growth, while Location B appears less suitable for vegetation.

8.6



- Diet B remains unimodal with a small right-hand tail.
- Most values sit in the middle classes (about 2 to 8 kg).
- Extreme weight losses occur rarely.
- Against Diet A, Diet B looks less consistent.
- The visual pattern implies Diet A is steadier, while Diet B's results spread out more.